

Vipin Sharma

Engineering Lead · 15+ Years · AI & Distributed Systems

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PROFESSIONAL SUMMARY

Engineering leader with 15+ years building distributed systems and production AI applications. Shipped LLM-powered products with measurable reliability, including output-validation loops, RAG-based cost optimization, and multi-agent workflows with human-in-the-loop checkpoints. Designed event-driven distributed platforms with async pipelines, shared validation contracts, and resilient external API integrations across Google and Microsoft Ads. Delivered business impact including 18% CPC improvement across 80% of campaigns, 30% lower overspending, and a new SKU targeting \$600K in monthly revenue. Led cross-functional teams of 6–8 engineers across the full product lifecycle.

PROFESSIONAL EXPERIENCE

Matchcraft

Software Engineering Team Lead

Santa Monica, CA
2021 – Present

AI-based Ad Planner and Ad Runner

Took an AI-native ad-generation platform from POC to production launch as a player-coach leading 7 engineers. Built on OpenAI APIs with a multi-agent pipeline and pgVector-backed RAG cache; launch targeted \$600K in monthly revenue.

- Designed an LLM self-review loop that validated output against platform constraints and regenerated on failure, achieving 80% first-pass compliance, 12% success on regeneration, and 8% abandonment.
- Shifted review from website-grounded to vertical-grounded scoring after outdated SMB sites caused false rejections, and introduced a vertical-classification agent that unlocked the RAG cache as a cost layer.
- Built a pgVector RAG cache keyed on vertical and services, storing assets with ad-engine performance metrics; cache hits reduced campaign generation from ~10 OpenAI calls to 2.
- Engineered an async fallback pipeline for generation failures that issued campaign IDs immediately, completed generation asynchronously, and notified users on completion, keeping production fallback rate under 5%.
- Architected a modular Java backend and Vue/Vite/Tailwind frontend with pluggable campaign engines, enabling other engineers to add 2 new ad engines and 4 campaign types without cross-team dependencies.

Category Subscription

Subscription-based platform delivering pre-built multilingual ad copy and keyword packages, with automated propagation of Master Category updates to live campaigns and ad engines.

- Productized a previously unmonetized internal capability into a paid SKU, driving a 3% revenue increase in year one through initial beta partner uptake.
- Owned the product end-to-end as sole engineering driver, replacing a fragile Excel-based taxonomy workflow with a structured create -> peer-review -> publish authoring portal.
- Built an automated propagation pipeline that fanned a single Master Category update to thousands of subscribed campaigns and to Google, cutting campaign setup time by 40%.

Bidding Algorithm

Led 3+ engineers in replacing a daily batch bidding system with a real-time, event-driven platform that fed hourly engine signals into a dedicated bidding service.

- Owned the architectural shift from batch to real-time, designing an event-driven platform that replaced day-late issue detection with near-real-time response to campaign performance.
 - Built an async delta-queue pipeline that consolidated per-campaign and per-keyword deltas into a single master-spend update per cycle, eliminating synchronous write contention.
 - Rewrote bidding to use hourly channel data instead of daily snapshots, improving CPC by 18% across ~80% of campaigns and cutting overspending by 30%.
 - Optimized report ingestion to process all campaigns across all engines in under an hour, enabling automatic pause-on-anomaly within an hour instead of a day later.
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Microsoft Ads Import

Microsoft-funded partnership delivering one-click campaign sync from Google to Microsoft Ads via the Microsoft Ads API with zero manual setup for clients.

- Built cross-platform sync using the Microsoft Ads API; adopted by 80% of Microsoft Ads clients and increased enrolled-client revenue by 22%.
- Architected Microsoft Ads campaign structure to mirror Google's, eliminating a parallel codebase and reducing Microsoft Ads update development time by over 40%.
- Preserved Google's optimized bid structure and ad copy on import; synced campaigns averaged 30% lower CPC and 20% higher CTR than manually created Microsoft Ads campaigns.
- Made sync resilient to unstable Microsoft Ads API jobs by designing a fire-and-forget polling pattern that persisted job IDs and polled to completion without blocking the daily cadence.

Overhauling Automated Task Processing

Redesigned the platform's distributed async task processing end-to-end, replacing per-engine processors with a shared processor that owned validation and enforced a uniform data-shape contract.

- Eliminated a class of cross-engine data corruption bugs by enforcing a shared data-shape contract before persistence, including cases where Google returned impression share as a fraction and Microsoft as a percentage.
- Decoupled fetch from process so each new ad engine required only a fetcher and shaping step, cutting development time by 50%, onboarding time by 65%, and eliminating 95%+ of regression risk.

Google Ads API Migration

- Rewrote the platform's Google Ads communication layer from scratch with zero client-facing interruption, including for thousands of clients managing campaigns directly via API.
- Selected as an early contributor in Google's Ads API beta program and provided structured feedback that was incorporated into the API design ahead of general availability.

Infosys

Senior Software Engineer

India
Sept 2008 – Dec 2014

- Led a team of 5+ engineers supporting 24x7 application availability, reducing outage incidents by 40% with no SLA breaches across 2 consecutive quarters.
- Designed a new NEFT transaction system that eliminated manual linking, and authored an SLA incident playbook adopted across teams for debugging and coordination.
- Delivered a new pharmacy claims screen that reduced time per patient by 25%, and designed an ePAO system that cleared a 6-month backlog of Payments & Accounts Office transactions.

EDUCATION

California State University, Long Beach

2016

Master of Science in Computer Science (GPA 4.0)

University of Pune, Pune, India

2008

Bachelor of Engineering in Computer Engineering

SKILLS

Technical Leadership: Cross-functional Lead (6–8 engineers), build/buy analysis, stakeholder management, mentoring, architectural design, sprint & roadmap planning

AI & LLM Engineering: Agent orchestration, multi-agent pipelines, RAG, vector databases (pgVector), self-review loops, hallucination reduction, output validation, OpenAI APIs, Anthropic APIs, Claude Code, AntiGravity (Agentic IDE), Ollama

Backend & Infrastructure: Java (Spring Boot), Python, SQL, C++, Microservices, AWS (EC2, RDS, Lambda, DynamoDB, EventBridge, SQS), Kafka, RabbitMQ, ClickHouse, PostgreSQL, External API Integrations (Google Ads, Microsoft Ads)

Frontend & Tools: TypeScript, JavaScript, Vue.js, Vite, Tailwind CSS, Docker, GitHub Actions, Cursor IDE